Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Bijak Minds
Project / Solution Name	Kak Nadiah – Orbyte Smart SME Copilot
Name (Leader)	Ammar Hakim Bin Dzulhaidi
Email & Phone (Leader)	bijakminds@gmail.com 016-404 6294
University Affiliation(s)	OPEN UNIVERSITY MALAYSIA
Live Prototype URL	https://590b068c-dd0e-4e78-9e04-78baeb87c670-00-3ttbi7c4xh6it.sisko.replit.dev/
Code Repository URL (public / access granted to judges)	https://github.com/bijakminds-cmd/kak-nadiah-orbyte-copilot

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Kak Nadiah – Orbyte Smart SME Copilot features a robust 3-tier end-to-end architecture tailored for micro and small-and-medium enterprises (MSMEs).

1. **Frontend Layer:** Built using HTML5 and Tailwind CSS structured around a Neo-Brutalism design system. It incorporates instant client-side i18n caching (11 core languages) paired with Gemini on-demand translation (supporting 249+ global languages), Chart.js real-time analytics, an interactive operations calendar, and locked-asset canvas layers for corporate logos and digital signatures.
2. **Backend Microservices:** Powered by a Node.js and Express.js REST API server managing secure session authentications, file uploads (Base64/PDF/Images), persistent JSON storage (`chat_history.json` and master company profile registries), and email dispatch.
3. **AI & Communication Layer:** Integrates Google Gemini 2.5 Flash API for multimodal document parsing and dynamic text generation, ElevenLabs Multilingual v2 API for natural text-to-speech audio synthesis, and Resend API for direct official letter emailing.
4. **Security & Governance:** Strict Human-in-the-Loop enforcement via RFC 6238 TOTP Google Authenticator 2FA (OTPLib) and simulated synchronization to the Orbyte Cloud ecosystem

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node.js (v20+) & Express.js (v4.x)	Backend runtime environment and REST API web server architecture	MIT License
Tailwind CSS (CDN)	Utility-first CSS framework for responsive Neo-Brutalism UI design	MIT License
Vanilla JavaScript (ES6+)	Frontend UI reactivity, DOM manipulation, client storage, and API fetch handling	MIT License

Name & Version	Purpose in Project	License
Chart.js (v4.x)	Real-time visual operational analytics and financial/document trend charting	MIT License
FontAwesome (v6.4)	Vector UI icons for navigation, document badges, and interactive controls	Font Awesome Free License
OTPLib (v12.x) QRCode (v1.5)	RFC 6238 TOTP generation and verification for employer 2FA security gate	MIT License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini 2.5 Flash API *(Google Cloud / AI Studio)*	Multimodal receipt & invoice vision analysis, auto-classification, discrepancy auditing, dynamic letter drafting, and 249+ language translation.	Sponsor-Provided / Free Tier
ElevenLabs Multilingual v2 API	Neural text-to-speech AI synthesis generating expressive Bahasa Melayu voice responses for Kak Nadiah Voice Copilot.	Developer / Paid Tier
Resend API	Transactional cloud email service for instant delivery of generated official business letters to Gmail/Outlook.	Free Tier
Google Fonts API	Typography loading for 'Plus Jakarta Sans' clean enterprise font styling.	Free Public API

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Cloud Platform	Containerized cloud environment for rapid prototype development, backend hosting, and live web preview execution.	Replit Core / Cloud Runner
Orbyte Cloud (AlphaRed)	Simulated enterprise backend synchronization target (Workspace ID: 416, Bijak Minds) for compliance audit logging.	Enterprise Sandbox Environment

Service & Provider	Purpose in Project	Plan / Region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Google Gemini 2.5 Flash	**Flash**Google Cloud / AI Studio	Pre-trained Multimodal LLM (Vision & Text Generation)	Google Cloud / AI Studio Terms of Service
ElevenLabs Multilingual v2	ElevenLabs	Pre-trained Neural Text-to-Speech (TTS) Voice Model	ElevenLabs Developer Terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Proprietary SME Operational Benchmark Registry	Self-collected / Internal mock dataset (Bijak Minds)	Private / Self-collected	Proprietary	**No** *(Simulated mock enterprise data only)*

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Google Gemini / Agentic AI Workspace	Assisting in rapid code structuring, Neo-Brutalism CSS styling guidelines, i18n dictionary translations, and proposal documentation drafting.	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every line of AI-assisted code, layout structure, and generated text was thoroughly reviewed, manually tested, and debugged by team leader Ammar Hakim. We maintained strict architectural oversight—ensuring API error handlers, fallback logic, port conflict resolution (`pkill -f node`), and security gates (2FA TOTP) were rigorously validated before integration. The team takes full responsibility for the final prototype behavior and compliance.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Google Cloud / SDC Organizers	Google Gemini 2.5 Flash API	Core intelligence backbone for multimodal invoice auditing, document classification, dynamic letter composition, and real-time UI translation.
AlphaRed / Orbyte	Orbyte Cloud Ecosystem & Guidelines	Benchmark workspace target (Organization ID: 416) for secure audit log synchronization and compliance verification.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The Google Gemini 2.5 Flash API is strictly essential to the core functionality of Kak Nadiah. It powers the multimodal reasoning required to parse unstructured vendor receipts, perform semantic invoice discrepancy detection, and execute dynamic multi-language translation. Replacing it would eliminate the project's primary value proposition: autonomous, accessible AI-powered compliance for non-technical SME operators.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Kak Nadiah adheres to strict privacy and data minimization principles. The application processes uploaded vendor invoices and company documents strictly on an ephemeral basis via secure Base64 payloads transmitted over HTTPS to the Google Gemini API. No sensitive customer personal identifiable information (PII) or permanent financial records are stored on third-party public clouds; persistent data is confined strictly to local server JSON files (`chat_history.json`, company profiles, and local browser `localStorage` for BYOK API keys). Furthermore, all privileged operations, such as financial ledger syncing or discrepancy approvals, are gated behind mandatory RFC 6238 TOTP Google Authenticator 2FA, ensuring complete human governance over automated workflows.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. Vision Parsing Rate Limits: Reliance on external LLM inference can introduce latency during peak API traffic, mitigated locally by structured JSON response schemas and robust fallback mockup data.

2. Handwritten Receipt OCR Limitations: Highly stylized or faded handwritten receipts may occasionally result in optical character recognition (OCR) parsing ambiguity; mitigated by providing direct human-in-the-loop manual editing fields across all generated documents and invoices.

3. Browser Speech Recognition Dependency: Voice Copilot features rely on the client browser's native Web Speech API support, which may vary across desktop and mobile browsers.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[25/10/2026]